

NAME

kh – two-player tactical space combat game command interface

SYNOPSIS

```
kh
kh --help
kh --version
kh --dbusr U --dbpass P --dbname D --port P
```

DESCRIPTION

Kepler's Horizon is a two-player tactical game of space combat where players build individualized ships, move them across a hex-based star map, and resolve combat without dice. The objective is to occupy opponent base star hexes and accumulate victory points.

This manual describes all available commands in the Kepler's Horizon command language as defined by the game grammar.

OPTIONS**Generic Program Information**

--help Output a usage message and exit.

-V, --version
Output the version number of **kh** and exit.

Database Connection

-U, --dbusr USER
Access the underlying database as *USER*

-D, --dbname DB_NAME
The name of the underlying database.

-P, --dbpass PASSWORD
The password for *USER*.

Database Connection

-T, --port PORT
The port number the server listens to for requests via REST. The default is 8080.

GAME OVERVIEW**Map Components****StarHex**

A map hex containing a large or small star (graph node)

BaseStarHex

A map hex containing a large star (6 total: 3 per player)

SpaceHex

A map hex without a star or warpline

WarpLine

Bi-directional edges connecting stars for instantaneous travel

Ship Types**WarpShip (W#)**

Ships capable of interstellar travel via warplines

SystemShip (S##)

Ships that cannot travel between stars unless carried by WarpShips

GAME SEQUENCE

Each player-turn consists of the following phases in order:

1. Count Victory Points
2. Build Ships (receive BP in Advanced scenario)
3. Movement
4. Resolve Combat
5. SystemShip Pick and Drop
6. End of Turn

CONCEPTS

Starting a Game

Games are started through the web interface, not via command line:

1. Portal

Login or create an account at the web portal.

2. Lobby

View open rooms or create a new one. Each room can host one game.

3. Room

Join a room and take a seat (A or B). When both seats are filled, either player can start the game.

4. Scenario Selection

The room creator selects the scenario:

learning

40 BP initial, WarpShips only, 1 base star, first to 1 VP wins

basic

50 BP initial, WarpShips and SystemShips, 1 base star, first to 2 VP wins

advanced

20 BP initial + 10 BP/turn, all rules active, 3 base stars, first to 3 VP wins

SCENARIOS COMMANDS

Commands are case-insensitive. Aliases are shown after commas.

Turn Management Commands

next, n Advance the active player to the next phase.

Syntax: next

Example:

> next

Phase advanced to: Movement

done, ZZ

Complete current player's turn and advance to opponent's first phase.

Syntax: done

Example:

> done

COMMAND: Your turn has ended. Standing down.

Information Commands

status Show game status and important records (current phase, turn number, BP, VP).

Syntax: status

fleet Show manifest of your fleet (all ships and their current attributes).

Syntax: fleet

Example:

> fleet

Your Fleet (Player A):

W1 The Vanguard [PD:5 B:3 S:2 T:1 M:6] at ARVEN

W2 The Sentinel [PD:4 B:2 S:3 T:0 M:0] at BELIX

crt Display the Combat Results Table

hex *location*

Show location and spatial information about the specified hex

online Show which players are online and their connection times

stats Display statistics about the game, your fleet, and opponent

fleet Show manifest of your fleet (all ships and their current attributes)

system *name*

Show environmental information about the specified star system

galaxy Complete overview of all systems and all ships in the game

score Show current game status: whose turn, phase, credits, tech level, and victory points

Economy Commands

extract scan

Scan current system for harvestable resources. Requires ship with LRS (Long Range Scanner).

extract *ship resource*

Extract resource from current system into ship's cargo hold. Requires TB (Transporter Beam) or DR (Drones).

Resources: FERROUS, RARE_EARTH, RADIOACTIVE, CRYSTALLINE, VOLATILE, WATER, ORGANIC, EXOTIC

market Display current market prices with trend indicators (↑² rising, ↑ stable, ↓¼ falling).

Prices fluctuate each round based on supply/demand and market forces.

market *resource*

Show price history for the specified resource over recent turns.

trade list

Show base trade prices for all resources.

trade buy *resource quantity*

Purchase resources at current market price, deducting credits.

trade sell *resource quantity*

Sell resources from cargo at 75% of market price.

trade transfer *from_ship to_ship resource quantity*

Move resources between ships at the same location.

fabricate list

Show available manufacturing recipes and their costs.

fabricate *recipe quantity*

Manufacture items from raw materials in cargo. Recipes include:

missiles

Basic missiles (4 per batch). Cost: 2 Fe, 1 Rd, 1 Vo. Time: 1 round.

tubes Tube upgrade (+1 capacity). Cost: 5 Fe, 3 Re, 2 Cr. Time: 3 rounds.

beams Beam upgrade (+1 rating). Cost: 8 Fe, 4 Re, 3 Cr. Time: 3 rounds.

screens Screen upgrade (+1 rating). Cost: 6 Fe, 2 Re, 4 Cr. Time: 3 rounds.

tech Tech research (+1 level). Cost: 10 Re, 5 Cr, 2 Ex. Time: 5 rounds.

salvage ship

Salvage wreckage at anomaly sites like The Graveyard (UMMA). Requires Drones (DR). Random outcome: 90% chance of recovering 2-5 units of random resources, 10% chance of hazard encounter dealing hull damage.

Build Commands

build, b

Show current build state (active drafts, available BP)

build new *code name*

bn *code name* Create a new ship draft with the specified code (e.g., W1, S01) and name

build drafts

bd List all pending build drafts

build drafts *ship_code*

bd *ship_code* Show detailed attributes of a specific draft ship

build set *attributes*, **bs** *attributes*

Set attributes on the current draft ship. Attributes are specified as:

pd=*N* or **d**=*N*

Power/Drive strength (1 BP per unit, provides movement and combat power)

b=*N* Beam weapon strength (1 BP per unit, max power for beam projector)

s=*N* Screen strength (1 BP per unit, absorbs hits equal to power + tech level)

t=*N* Number of missile tubes (1 BP per tube, requires 1 PD to fire)

m=*N* Number of missiles (1 BP per 3 missiles, each does 2 + tech level hits)

sr=*N* SystemShip racks (1 BP per rack, WarpShips only, carries 1 SystemShip per rack)

Example: build set pd=7 b=3 s=2 t=1 m=6

Note: Warp Generator costs 5 BP (automatically added for WarpShips)

build commit

bc Commit the current draft, spending BP and adding ship to your fleet

build cancel

bx Cancel the current draft without spending BP

Deployment Commands

deploy *ship_code system_name*

ds *ship_code system_name* Deploy the specified ship to the specified base star hex (must be your base star)

Movement Commands

move *ship_code destination*

m *ship_code destination* Move the specified ship to destination, optionally through waypoints. Movement costs:

- 1 PD: Move to adjacent hex

- 1 PD: Enter warpline at star hex
- 1 PD: Travel full warpline to destination star
- 1 PD: Pick up one SystemShip
- 1 PD: Drop off one SystemShip

Ships must stop at star hexes occupied by enemy ships.

Ships may freely move through space hexes with enemy ships.

Combat Commands

combat

c Show status of current combat situation

combat order *ship tactic target allocation*

co *ship tactic target allocation* Issue combat orders for a ship. Components:

ship Your ship identifier (e.g., W1, S05)

tactic Combat tactic: **attack**, **dodge**, or **escape** (WarpShips only)

target Enemy ship identifier to target

allocation

Power allocation from PD to attributes:

pd=*N* or **d**=*N*

Drive allocation (for maneuver, affects CRT row/column)

b=*N* Beam power (cannot exceed built capacity)

s=*N* Screen power (cannot exceed built capacity)

t=*N* Tubes to power (1 PD per tube)

m=*N* Missiles to fire (requires powered tubes)

Note: Total allocation cannot exceed current PD

Note: Cannot use Beams/Screens and Missiles in same round

combat commit

cc Commit all combat orders for this round

combat cancel

cx Cancel all combat orders and re-issue

combat drafts

cd List pending combat orders

combat apply *ship damage_allocation*

ca *ship damage_allocation* Apply damage hits to ship attributes (defender chooses where hits are taken)

Pick and Drop Commands

pick *systemship warship*

p *systemship warship* Pick up the specified SystemShip into the specified WarpShip's rack

drop *systemship warship*

dd *systemship warship* Drop the specified SystemShip from the specified WarpShip at current location

Repair and Resupply Commands

repair *ship_code attribute N*

rp *ship_code attribute N* Repair the specified ship using BP (1 BP repairs 1 unit of any attribute up to original max). Must be at your base star hex.

resupply *ship_code quantity*

rs *ship_code quantity* Resupply the specified ship with missiles (1 BP per 3 missiles, up to original max)

Help Commands

help ? Show help screen and list all help topics

SHIP ATTRIBUTES

Power/Drive (PD)

Engine strength. Provides movement points and combat power allocation. Cost: 1 BP per unit.

Warp Generator (WG)

Enables interstellar travel and warpline usage. Cost: 5 BP. Required for WarpShips.

Beams (B)

Energy weapon projector. Max strength = BP spent. Cost: 1 BP per unit. Damage = power + tech level.

Screens (S)

Protective energy screen. Absorbs hits = power + tech level. Cost: 1 BP per unit.

Tubes (T)

Missile launchers. Each tube fires 1 missile/round. Requires 1 PD to power. Cost: 1 BP per tube.

Missiles (M)

Projectile weapons. Damage = 2 + tech level. Cost: 1 BP per 3 missiles.

SystemShip Racks (SR)

Carries SystemShips (WarpShips only). 1 SystemShip per rack. Cost: 1 CR per rack.

Extraction Equipment

Long Range Scanner (LRS)

Detects resources in current system. Required for extract scan. Cost: 3 CR per unit.

Transporter Beam (TB)

Extracts resources from standard locations. Required for extract. Cost: 5 CR per unit.

Drones (DR)

Extracts resources from hazardous locations. Required for extract in dangerous zones. Cost: 4 CR per unit.

Note: LRS, TB, and DR do not affect ship destruction. Only PD, B, S, T, M, SR reaching zero destroys the ship.

TECHNOLOGY LEVELS

In Advanced scenario, tech level increases every 4 turns:

Turns 1-4: Level 0

Turns 5-8: Level 1

Turns 9-12: Level 2

Turns 13-16: Level 3

(continues +1 per 4 turns)

Tech level adds to:

- Beam damage when hitting
- Missile damage when hitting (2 + tech level)
- Screen absorption when powered

COMBAT RESOLUTION

Combat Results Table (CRT)

The CRT determines hit/miss based on:

- Firing ship's tactic (Attack/Dodge/Retreat)
- Target ship's tactic (Attack/Dodge/Retreat)
- Drive difference (Firing Drive - Target Drive)

Combat Tactics

Attack Aggressive offense. Best hit chance when drive advantage is 0 to +2.

Dodge Defensive maneuvering. Good hit chance at +1/+2 drive, hard to hit at 0/-1.

Retreat (WarpShips only)

Attempt to escape combat. Must get "Escapes" result against all enemy ships.

Damage Application

1. Calculate hits from all weapons that hit
2. Subtract Screen power + tech level (if Screen powered)
3. Apply remaining "effective hits" to ship attributes (owner chooses where)
4. Ship destroyed when all attributes except Warp Generator reduced to 0

Special Combat Rules

- Beams and Screens can be used together
- Missiles cannot be used same round as Beams/Screens
- Missiles always use Attack tactic with their own drive setting
- SystemShips cannot Retreat (must be carried by WarpShip)
- Combat ends when: all ships of one side destroyed/retreated, or 3 consecutive rounds with no unabsorbed damage (attacker must withdraw)

MOVEMENT RULES

1. WarpShips must stop at star hexes occupied by enemy ships
2. WarpShips may move through space hexes with enemy ships
3. WarpLines are treated as space hexes for movement
4. Ships with PD=0 cannot move
5. PD used for movement is not expended (full PD available for combat)
6. Cannot move onto enemy base star hex during turn 1

VICTORY CONDITIONS

Dominance Victory

Accumulate required victory points (1/2/3 depending on scenario) by occupying enemy base stars at start of your turn

Elimination Victory

Destroy all enemy ships

Draw Neither player has effective ships remaining (Learning/Basic scenarios only)

BUILD POINT COSTS

Attribute	Cost
Power/Drive (PD)	1 BP per unit
Warp Generator (WG)	5 BP
Beam (B)	1 BP per unit
Screen (S)	1 BP per unit
Tube (T)	1 BP per tube
Missiles (M)	1 BP per 3 missiles
SystemShip Rack (SR)	1 BP per rack

EXAMPLES**Building a WarpShip**

> **build new** W2 Destroyer
 > **build set** pd=5 b=3 s=2 t=1 m=3
 > **build commit**

The same commands with aliased versions:

> **bn** w Falcon
 > **bs** pd=5 b=3 s=2 t=1 m=3
 > **bc**

Note: If the player doesn't care, the game will generate a numeric ID. Entering W2 is not necessary -- unless you want your ship numbered W2.

Total cost: 5+3+2+1+1+5(WG) = 17 BP

Tech level 0 (if built turns 1-4)

Building a SystemShip Fighter

> **build new** S55 Fighter
 > **build set** pd=1 t=1 m=6
 > **build commit**

Or use:

> **bn** S55 Avenger
 > **bs** pd=1 t=1 m=6
 > **bc**

Total cost: 1+1+2 = 4 BP

No Warp Generator (SystemShip)

Movement Example

> **move** W6 Girsu Kish

Or the short hand:

> **m** W6 Girsu Kish

W6 moves from current hex to Girsu (1 PD), along warpline to next star (1 PD), then 3 hexes to Kish (3 PD) = 5 PD total

Combat Order Example

> **combat order W3 attack S25 d=0 b=3 s=2**

Or use with same result:

> **co W3 a S25 d=0 b=3 s=2**

W3 attacks S25, allocating 0 to drive, 3 to beam, 2 to screen (requires PD¥5)

Missile Fire Example

> **combat order S25 dodge W3 d=4 t=1 m=1**

> (missile drive setting: 3)

Or use:

> **co S25 d W3 d=4 t=1 m=1**

> (missile drive setting: 3)

S25 dodges while firing 1 missile at W3 with drive=3 (requires PD¥5)

CONSTRAINTS AND INVARIANTS

- No randomness - all outcomes deterministic
- No hidden information - all game state is public
- Rules do not change mid-game
- Equal inputs yield equal outcomes
- Ships retain original tech level permanently
- Warp Generators never take damage (ship destroyed when all else is 0)
- SystemShips cannot have Warp Generators or SystemShip Racks
- WarpShips cannot be carried in SystemShip Racks

REPAIR AND RESUPPLY

- Ship must start turn at your base star hex
- SystemShips can be repaired/resupplied while loaded on WarpShip
- 1 BP repairs 1 unit of any attribute
- Cannot repair beyond original built capacity
- Tech level never changes
- 1 BP resupplies up to 3 missiles
- Cannot resupply beyond original missile capacity

SYSTEMSHIP PICKUP/DROP**During Movement**

- Costs 1 PD per SystemShip picked up or dropped
- Can be done at any star hex during movement

During Combat

- Requires: Drive=0, Screen=0, Dodge or Retreat tactic
- Only 1 SystemShip per combat round
- May fire Beam (but not Missiles) while picking up/dropping

- Dropped SystemShips cannot fire that round
- Picked up SystemShips cannot fire but may power Screens

After Combat

- Free rearrangement of SystemShips at star hexes
- No movement cost
- Active player only

FILES

~/kh/saves/

Directory containing saved game files

~/kh/lastgame

Automatically saved game when quitting

SEE ALSO

RULES.md(5), **COMBAT_RULES.md(5)**, **DESIGN.md(5)**

NOTES

- All ship records must be maintained by players
- Records are shown to opponent after (not during) game
- In Advanced scenario, track BP received and expended
- Combat orders are written simultaneously and revealed together
- Fractions of BP from missile resupply are not saved

THE GRIMOIRE

The Grimoire of Stars is the compendium of known space, containing detailed information about each star system. Knowledge levels range from Unknown (name only) to Intimate (complete secrets).

Systems are organized by territory:

- **Western Alliance:** ARVEN (capital), BELIX, CAYRU
- **Eastern Powers:** ZAREK, ASTREX, BRION
- **Contested Reach:** KORAL, QUELL, SYDRA, VARYN, WEXAR, XYLEN, TAVON, and others

Each system entry includes celestial bodies, notable moons, resources, populations, infrastructure, and anomalies.

RESOURCES

Eight types of raw materials can be extracted and traded:

Resource	Abbrev	Primary Uses
FERROUS	Fe	Hull construction
RARE_EARTH	Re	Electronics, screens
RADIOACTIVE	Rd	Power, weapons
CRYSTALLINE	Cr	Beams, warp drives
VOLATILE	Vo	Fuel, missiles
WATER	H2O	Life support
ORGANIC	Or	Food, medicine
EXOTIC	Ex	Special technology

MARKET DYNAMICS

The commodity market fluctuates based on supply and demand:

- Prices update each round after both players complete turns
- Buying increases demand, pushing prices up
- Selling increases supply, pushing prices down
- Random market forces also affect prices
- Prices clamped to 50%-200% of base value

Use **market** to view current prices and trends, then **trade buy/sell** to act.

FACILITIES

Facilities at star systems provide strategic advantages to the controlling player. Control is established by occupying a system with no enemy ships for 2 consecutive turns.

Facility	Effect
SHIPYARD	Build new ships
REPAIR_DOCK	Repair damaged ships away from base
REFINERY	Fabrication bonuses
TRADE_HUB	+5 CR income per turn
FORTRESS	Combat defensive bonus

Use **repair** at captured REPAIR_DOCKs or SHIPYARDs to repair ships in the field.

BUGS

Report bugs to the Kepler's Horizon issue tracker.

AUTHORS

Game design and implementation by sibomots.

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GAME DESIGN PHILOSOPHY

Kepler's Horizon is intentionally minimal. Strategic depth emerges from:

- Constrained movement
- Simultaneous decision-making
- Irreversible consequences

The rules favor clarity over complexity.